DATA STRUCTURE
LAB PROGRAM

25. Implementation of Minimum Spanning Tree using Kruskal Algorithm

```
#include <stdio.h>

#define MAX 20

struct Edge
{
   int u, v, weight;
};

int parent[MAX];

int find(int i)
{
   while (parent[i] != i)
      i = parent[i];
   return i;
}
```

```c
void unionSet(int a, int b)
{
    parent[a] = b;
}

int main()
{
    int n, e, i, j;

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter number of edges: ");
    scanf("%d", &e);

    struct Edge edge[e], temp;

    printf("Enter edges (source destination weight):\n");
    for(i = 0; i < e; i++)
    {
        scanf("%d %d %d",
              &edge[i].u,
              &edge[i].v,
```

```
        &edge[i].weight);
}

/* Sort edges by weight */
for(i = 0; i < e - 1; i++)
{
    for(j = 0; j < e - i - 1; j++)
    {
        if(edge[j].weight > edge[j + 1].weight)
        {
            temp = edge[j];
            edge[j] = edge[j + 1];
            edge[j + 1] = temp;
        }
    }
}

for(i = 0; i < n; i++)
    parent[i] = i;

printf("\nEdges in MST:\n");

int count = 0, minCost = 0;
```

```
    for(i = 0; i < e && count < n - 1; i++)
    {
        int u = find(edge[i].u);
        int v = find(edge[i].v);

        if(u != v)
        {
            printf("%d -- %d = %d\n",
                   edge[i].u,
                   edge[i].v,
                   edge[i].weight);

            minCost += edge[i].weight;
            unionSet(u, v);
            count++;
        }
    }

    printf("\nTotal Cost of MST = %d\n", minCost);

    return 0;
}
```

Output

```
Enter number of vertices: 4
Enter number of edges: 5
Enter edges (source destination weight):
1 2 4
5 42 45
5 4 2
5 5 7
2 4 88

Edges in MST:
1 -- 2 = 4
2 -- 4 = 88

Total Cost of MST = 92


=== Code Execution Successful ===
```